4.1.4 Production, costs and revenue

4.1.4.1 Production and productivity

Content	Additional information
• Production converts inputs, or the services of factors of production such as capital and labour, into final output. • The meaning of productivity, including labour productivity.	

4.1.4.2 Specialisation, division of labour and exchange

Content	Additional information
• The benefits of specialisation and division of labour. • Why specialisation necessitates an efficient means of exchanging goods and services, such as the use of money as a medium of exchange.	

4.1.4.3 The law of diminishing returns and returns to scale

Content	Additional information
• The difference between the short run and the long run. • The difference between marginal, average and total returns. • The law of diminishing returns. • Returns to scale. • The difference between increasing, constant and decreasing returns to scale.	Students should appreciate that both the law of diminishing returns and returns to scale explain relationships between inputs and output. They should also understand that these relationships have implications for costs of production.

4.1.4.4 Costs of production

Content	Additional information
• The difference between fixed and variable costs. • The difference between marginal, average and total costs. • The difference between short-run and long-run costs. • The reasons for the shape of the marginal, average and total cost curves. • How factor prices and productivity affect firms' costs of production and their choice of factor inputs.	Students should be able to calculate different costs from given data. They should also be able to draw and interpret cost curves.

4.1.4.5 Economies and diseconomies of scale

Content	Additional information
• The difference between internal and external economies of scale. • Reasons for diseconomies of scale. • The relationship between returns to scale and economies or diseconomies of scale. • The relationship between economies of scale, diseconomies of scale and the shape of the long-run average cost curve. • The L-shaped long-run average cost curve. • The concept of the minimum efficient scale of production.	Students should be able to categorise and give examples of both internal and external economies of scale. Students should understand the significance of the minimum efficient scale for the structure of an industry and barriers to entry.

4.1.4.6 Marginal, average and total revenue

Content	Additional information
• The difference between marginal, average and total revenue. • Why the average revenue curve is the firm's demand curve. • The relationship between average and marginal revenue. • The relationship between marginal revenue and total revenue.	Students should be able to calculate marginal, average and total revenue from given data. They should also be able to draw and interpret revenue curves.

4.1.4.7 Profit

Content	Additional information
• Profit is the difference between total revenue and total costs. • The difference between normal and abnormal (supernormal) profit. • The role of profit in a market economy.	

4.1.4.8 Technological change

Content	Additional information
• The difference between invention and innovation. • Technological change can affect methods of production, productivity, efficiency and firms' costs of production. • Technological change can lead to the development of new products, the development of new markets and may destroy existing markets. • Technological change can influence the structure of markets.	Students should understand how the process of creative destruction is linked to technological change.